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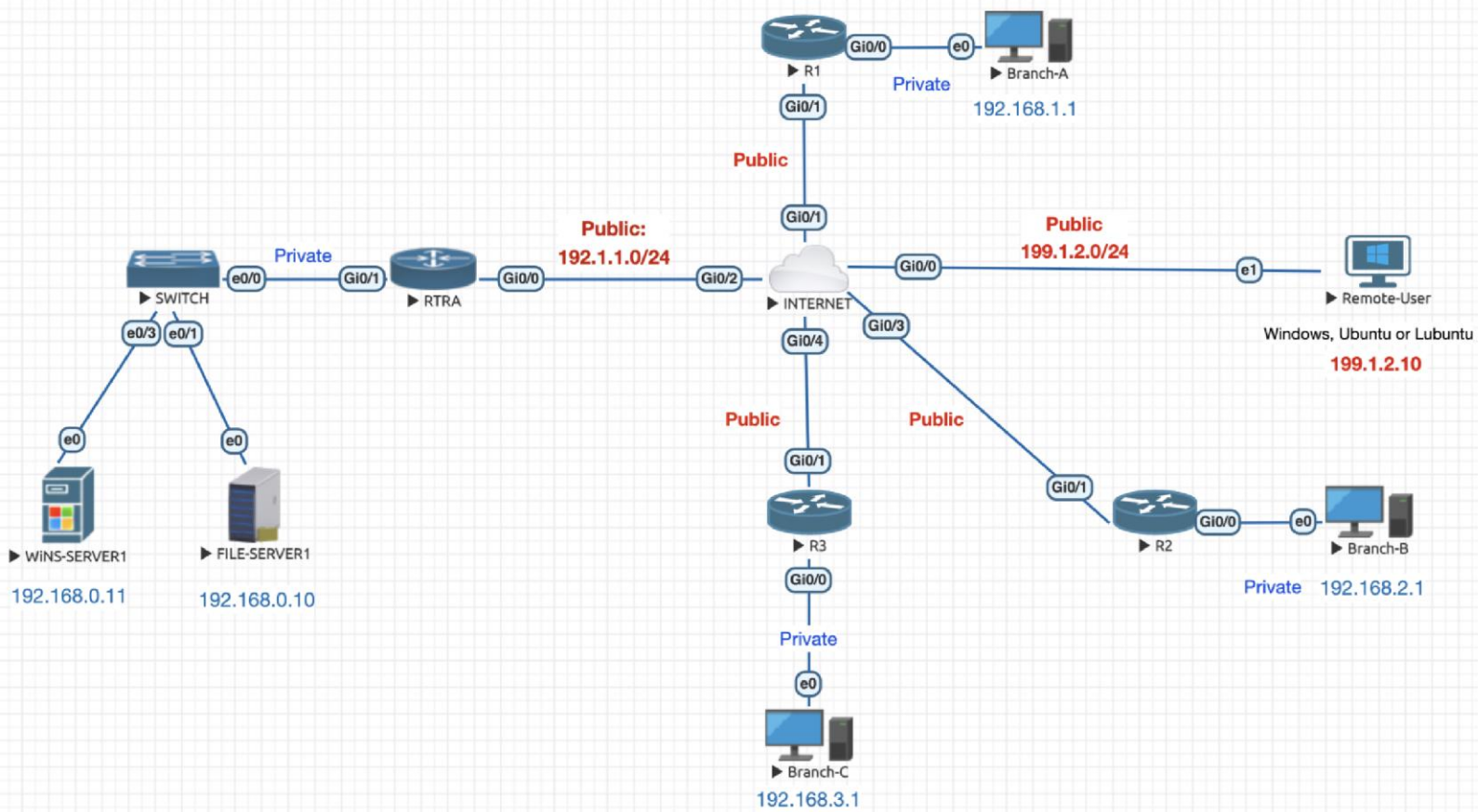
Introduction to Enterprise Networks

Spring 2025

Lab 6

IPSEC - Tunneling – DMVPN

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Lab Instructions:

Before connecting any routers, ensure that the IOS version on each one supports encryption. This can be done by running the following command:

```
Router# show version
```

- ** The IOS name in the output must contain “K9”.
- ** You can save and clear the configs after each objective if you want to.
- ** You can use the following commands to verify your configurations.

- **show crypto session**
- **show crypto ipsec sa**
- **show crypto isakmp sa**
- **show crypto map**
- **show crypto session remote ip-address detail**

Objective 1: Site to Site VPN

Create loopbacks on R1, R2 and R3 with desired IP addresses:

Any traffic from Branches A, B and C to the Fileserver must be encrypted via IPSEC

1. Create IPSec tunnels between R1, R2 and RTRA. Configure in such a way that when any of the branches ping the FileServer1 the data is encrypted. Use Wireshark capture.

Paste Config + screenshots showing the encrypted traffic.

Device configurations for RTRA (R4), R1, and R2:

```
R4 config:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key admin address 221.1.2.2
crypto isakmp key admin2 address 211.1.2.2
!
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto map R1_cryptomap 10 ipsec-isakmp
set peer 221.1.2.2
set transform-set MYTRANSFORMSET
```

```
match address 100
!
crypto map R1_cryptomap 20 ipsec-isakmp
set peer 211.1.2.2
set transform-set MYTRANSFORMSET
match address 101
!
interface GigabitEthernet0/1
ip address 192.1.1.2 255.255.255.0
crypto map R1_cryptomap
!
access-list 100 permit ip host 192.168.0.11 host 192.168.1.1
access-list 101 permit ip host 192.168.0.11 host 192.168.2.1

R1 config:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key admin address 192.1.1.2
!
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto map R4_cryptomap 10 ipsec-isakmp
set peer 192.1.1.2
set transform-set MYTRANSFORMSET
match address 100
!
interface GigabitEthernet0/1
ip address 221.1.2.2 255.255.255.0
crypto map R4_cryptomap
!
access-list 100 permit ip host 192.168.1.1 host 192.168.0.11

R2 config:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key admin2 address 192.1.1.2
!
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto map R4_cryptomap 10 ipsec-isakmp
set peer 192.1.1.2
set transform-set MYTRANSFORMSET
match address 100
!
interface GigabitEthernet0/1
crypto map R4_cryptomap
!
access-list 100 permit ip host 192.168.2.1 host 192.168.0.11
```

Wireshark captures of Both R1 and R2 loopbacks pinging the file server:

Top Wireshark Capture: Packet List

No.	Time	Source	Destination	Protocol	Length	Info
4	5.597688	192.1.1.2	221.1.1.2	ESP	182	ESP (SPI=0x3406f116)
5	5.598641	221.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x0806e23b)
6	5.599576	192.1.1.2	221.1.1.2	ESP	182	ESP (SPI=0x3406f116)
7	5.600171	221.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x0806e23b)
8	5.600811	192.1.1.2	221.1.1.2	ESP	182	ESP (SPI=0x3406f116)
9	5.601390	221.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x0806e23b)
10	5.601892	192.1.1.2	221.1.1.2	ESP	182	ESP (SPI=0x3406f116)
11	5.602420	221.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x0806e23b)
12	5.602955	192.1.1.2	221.1.1.2	ESP	182	ESP (SPI=0x3406f116)
13	5.603474	221.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x0806e23b)
14	8.176377	192.1.1.2	211.1.1.2	ESP	182	ESP (SPI=0x3b0ba22)
15	8.177348	211.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x52929867)
16	8.178120	192.1.1.2	211.1.1.2	ESP	182	ESP (SPI=0x3b0ba22)
17	8.178738	211.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x52929867)
18	8.179357	192.1.1.2	211.1.1.2	ESP	182	ESP (SPI=0x3b0ba22)
19	8.179875	211.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x52929867)
20	8.180448	192.1.1.2	211.1.1.2	ESP	182	ESP (SPI=0x3b0ba22)
21	8.180990	211.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x52929867)
22	8.181577	192.1.1.2	211.1.1.2	ESP	182	ESP (SPI=0x3b0ba22)
23	8.182144	211.1.1.2	192.1.1.2	ESP	182	ESP (SPI=0x52929867)

Bottom Wireshark Capture: Packet Details

Frame 23: 182 bytes on wire (1456 bits), 182 bytes captured (1456 bits) on interface \Device\NPF_{25513793-3101-4A3E-B982-F80D39A80BDC}, id 0

- Ethernet II, Src: Cisco_F7:86:c5 (00:17:5a:f7:86:c5), Dst: Cisco_4f:cd:61 (00:23:04:4f:cd:61)
 - Destination: Cisco_4f:cd:61 (00:23:04:4f:cd:61)
 - Source: Cisco_F7:86:c5 (00:17:5a:f7:86:c5)
 - Type: IPv4 (0x0800)
- Internet Protocol Version 4, Src: 211.1.1.2, Dst: 192.1.1.2
 - 0100 = Version: 4
 - 0101 = Header Length: 20 bytes (5)
 - Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
 - Total Length: 168
 - Identification: 0x0400 (1024)
 - Flags: 0x00
 - Fragment Offset: 0
 - Time to Live: 254
 - Protocol: Encap Security Payload (50)
 - Header checksum status: Unverified
 - Source Address: 211.1.1.2
 - Destination Address: 192.1.1.2
- Encapsulating Security Payload
 - ESP SPI: 0x52929867 (138538983)
 - ESP Sequence: 5

Packet Bytes

```

0000  00 23 04 4f cd 61 80 17 5a f7 86 c5 00 00 45 00  #O.8...E
0010  00 a8 04 00 00 00 fe 32 22 1d d3 01 02 02 c0 01  ....2".....
  
```

Show commands:

```
R1>en
R1#show crypto ipsec sa

interface: GigabitEthernet0/1
  Crypto map tag: R4_cryptomap, local addr 221.1.2.2

protected vrf: (none)
local  ident (addr/mask/prot/port): (192.168.1.1/255.255.255.255/0/0)
remote ident (addr/mask/prot/port): (192.168.0.11/255.255.255.255/0/0)
current_peer 192.1.1.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 79, #pkts encrypt: 79, #pkts digest: 79
  #pkts decaps: 71, #pkts decrypt: 71, #pkts verify: 71
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 1, #recv errors 0

  local crypto endpt.: 221.1.2.2, remote crypto endpt.: 192.1.1.2
  path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
  current outbound spi: 0x0(0)
  PFS (Y/N): N, DH group: none

inbound esp sas:

inbound ah sas:

inbound pcp sas:

outbound esp sas:

outbound ah sas:

outbound pcp sas:
```

```
R1#
```

```

R4>en
R4#show crypto ipsec sa

interface: GigabitEthernet0/1
  Crypto map tag: R1_cryptomap, local addr 192.1.1.2

protected vrf: (none)
local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
current_peer 201.1.2.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 7115, #pkts encrypt: 7115, #pkts digest: 7115
  #pkts decaps: 7093, #pkts decrypt: 7093, #pkts verify: 7093
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 1, #recv errors 0

local crypto endpt.: 192.1.1.2, remote crypto endpt.: 201.1.2.2
path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
current outbound spi: 0xEF77E46A(4017611882)
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0x171653E4(387339236)
    transform: esp-aes esp-sha-hmac ,
    in use settings = {Tunnel, }
    conn id: 2061, flow_id: Onboard VPN:61, sibling_flags 80000046, crypto map: R1_cryptomap
    sa timing: remaining key lifetime (k/sec): (4494446/1691)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0xEF77E46A(4017611882)
    transform: esp-aes esp-sha-hmac ,
    in use settings = {Tunnel, }
    conn id: 2062, flow_id: Onboard VPN:62, sibling_flags 80000046, crypto map: R1_cryptomap
    sa timing: remaining key lifetime (k/sec): (4494446/1691)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE

```

2. Create a GRE-IPSEC tunnel between R3 and RTRA. Traffic between Branch-C and the **Wins-Server** should go encrypted via the GRE tunnel interface.

Attach a screenshot showing that traffic is going encrypted through the GRE tunnel. +
The commands used for this part.

IPSEC configuration on RTRA (R4) and R3:

Note* isakmp and ipsec was already configured when setting up GRE:

```

R4 Config:
access-list 102 permit gre any any
interface Tunnel1

```

```
ip address 10.10.10.1 255.255.255.0
tunnel source 192.1.1.2
tunnel destination 201.1.2.2
!
crypto map R1_cryptomap 30 ipsec-isakmp
set peer 201.1.2.2
set transform-set MYTRANSFORMSET
match address 102
!
crypto isakmp key admin3 address 201.1.2.2

R3 config:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key admin3 address 192.1.1.2
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto map R4_cryptomap 10 ipsec-isakmp
set peer 192.1.1.2
set transform-set MYTRANSFORMSET
match address 100
!
interface GigabitEthernet0/1
crypto map R4_cryptomap
!
access-list 100 permit gre any any
interface Tunnel1
ip address 10.10.10.2 255.255.255.0
tunnel source 201.1.2.2
tunnel destination 192.1.1.2
!
```

Wireshark capture showing connectivity:

Capturing from Intel(R) 82577LM Gigabit Network Connection: Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

esp

No.	Time	Source	Destination	Protocol	Length	Info
221	103.324327	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)
225	112.089845	192.1.1.2	201.1.2.2	ESP	182	ESP (SPI=0xbd0e65cd)
227	113.112384	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)
233	121.649915	192.1.1.2	201.1.2.2	ESP	182	ESP (SPI=0xbd0e65cd)
234	122.912449	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)
241	131.173955	192.1.1.2	201.1.2.2	ESP	182	ESP (SPI=0xbd0e65cd)
242	132.308293	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)
246	140.814001	192.1.1.2	201.1.2.2	ESP	182	ESP (SPI=0xbd0e65cd)
247	141.328386	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)
254	149.906007	192.1.1.2	201.1.2.2	ESP	182	ESP (SPI=0xbd0e65cd)
255	150.984331	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)
263	159.370046	192.1.1.2	201.1.2.2	ESP	182	ESP (SPI=0xbd0e65cd)
264	160.600518	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)
268	168.698076	192.1.1.2	201.1.2.2	ESP	182	ESP (SPI=0xbd0e65cd)
269	170.332694	201.1.2.2	192.1.1.2	ESP	182	ESP (SPI=0xdc747f30)

< >

> Frame 1: 182 bytes on wire (1456 bits), 182 bytes captured (1456 bits) on interface \Device\NPF_{25513793-3101-4A3E-B982-FB0D39ABDBCD}, id 0
> Ethernet II, Src: Cisco_f7:86:c5 (00:17:5a:f7:86:c5), Dst: Cisco_4f:cd:61 (00:23:04:4f:cd:61)
> Internet Protocol Version 4, Src: 201.1.2.2, Dst: 192.1.1.2
> Encapsulating Security Payload

0000 00 23 04 4f cd 61 00 17 5a f7 86 c5 08 00 45 c0 #.O.a..Z.....E.

Intel(R) 82577LM Gigabit Network Connection: Ethernet: <live capture in progress> | Packets: 269 · Displayed: 57 (21.2%) | Profile: Default

```

R3>
R3>en
R3#show crypto ipsec sa

interface: GigabitEthernet0/1
  Crypto map tag: R4_cryptomap, local addr 201.1.2.2

  protected vrf: (none)
  local ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
  remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
  current_peer 192.1.1.2 port 500
    PERMIT, flags={origin_is_acl,}
    #pkts encaps: 7103, #pkts encrypt: 7103, #pkts digest: 7103
    #pkts decaps: 7125, #pkts decrypt: 7125, #pkts verify: 7125
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

  local crypto endpt.: 201.1.2.2, remote crypto endpt.: 192.1.1.2
  path mtu 1500, ip mtu 1500, ip mtu idb GigabitEthernet0/1
  current outbound spi: 0x171653E4(387339236)
  PFS (Y/N): N, DH group: none

  inbound esp sas:
    spi: 0xEF77E46A(4017611882)
      transform: esp-aes esp-sha-hmac ,
      in use settings = {Tunnel, }
      conn id: 2041, flow_id: Onboard VPN:41, sibling_flags 80000046, crypto map: R4_cryptomap
      sa timing: remaining key lifetime (k/sec): (4395436/1598)
      IV size: 16 bytes
      replay detection support: Y
      Status: ACTIVE

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:
    spi: 0x171653E4(387339236)
      transform: esp-aes esp-sha-hmac ,
      in use settings = {Tunnel, }
      conn id: 2042, flow_id: Onboard VPN:42, sibling_flags 80000046, crypto map: R4_cryptomap
      sa timing: remaining key lifetime (k/sec): (4395436/1598)
      IV size: 16 bytes
      replay detection support: Y
      Status: ACTIVE

  outbound ah sas:

  outbound pcp sas:

```

R3#

Objective 2: Remote VPN

In this part you need to create a “Remote VPN” or “Easy VPN”. The remote user needs to access the Wins-Server and the Fileserver. Remote user is a windows VM that has a Cisco VPN client installed on it or a client VPN such as OpenVPN installed and configured on a lightweight Ubuntu for the virtual users.

Configure Easy VPN on RTRA and use a client VPN on the remote user. The remote user should be able to access both servers via a VPN connection to RTRA.

*** EXTRA CREDIT:

Create two different groups for accessing each server. “Users” group should only access the Fileserver with their own credentials (username and password) and “admins” group should only access the Fileserver with different credentials. Use isakmp client configuration group, different pools and aaa authentication to fulfill this requirement.

Paste screenshots showing successful VPN connection from the remote user to the Servers via RTRA

Paste screenshots showing the data is encrypted.

I used the built in Cisco IPSEC VPN tool on the MacBook Pro to connect to the VPN of RTRA. Here is the configuration:

The screenshot shows the Cisco IPsec configuration utility. The left sidebar has 'Cisco IPsec' selected, with 'DNS' and 'Proxies' as sub-options. The main configuration area is divided into sections. The top section contains 'Display name' (VPN), 'Server address' (192.1.1.2), 'Account name' (VPNUSER), and 'Password'. The 'Authentication' section below it contains 'Type' (Shared secret), 'Shared secret' (masked with dots), and 'Group name' (VPNGROUP). At the bottom, there are three buttons: 'Remove Configuration...', 'Cancel', and 'OK'.

Wireshark capture of VPN working:

ip addr == 192.1.1.2

No.	Time	Source	Destination	Protocol	Length	Info
201	178.409295	192.1.1.3	192.1.1.2	ISAKMP	806	Aggressive
202	178.411100	192.1.1.2	192.1.1.3	ISAKMP	246	Informational
204	178.419778	192.1.1.3	192.1.1.2	ISAKMP	806	Aggressive
205	178.440956	192.1.1.2	192.1.1.3	ISAKMP	458	Aggressive
206	178.464775	192.1.1.3	192.1.1.2	ISAKMP	182	Aggressive
207	178.466518	192.1.1.2	192.1.1.3	ISAKMP	134	Informational
208	178.467150	192.1.1.2	192.1.1.3	ISAKMP	118	Transaction (Config Mode)
209	180.236492	192.1.1.2	224.0.0.5	OSPF	94	Hello Packet
215	183.953199	192.1.1.3	192.1.1.2	ISAKMP	134	Transaction (Config Mode)
216	183.955009	192.1.1.2	192.1.1.3	ISAKMP	118	Transaction (Config Mode)
217	183.955273	192.1.1.3	192.1.1.2	ISAKMP	118	Transaction (Config Mode)
218	183.955570	192.1.1.3	192.1.1.2	ISAKMP	214	Transaction (Config Mode)
219	183.964086	192.1.1.2	192.1.1.3	ISAKMP	438	Transaction (Config Mode)
220	184.004562	192.1.1.3	192.1.1.2	ISAKMP	422	Quick Mode
221	184.007800	192.1.1.2	192.1.1.3	ISAKMP	230	Quick Mode
222	184.027752	192.1.1.3	192.1.1.2	ISAKMP	182	Quick Mode
223	184.087670	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
224	184.087913	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
225	184.094467	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
226	184.094509	192.1.1.3	192.1.1.2	ESP	134	ESP (SPI=0xf6354fca)
227	184.094538	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
228	184.099173	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
229	184.099218	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
230	184.105572	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
231	184.105848	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
232	184.112972	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
235	184.149225	192.1.1.3	192.1.1.2	ESP	134	ESP (SPI=0xf6354fca)
236	184.149265	192.1.1.3	192.1.1.2	ESP	134	ESP (SPI=0xf6354fca)
237	184.184561	192.1.1.3	192.1.1.2	ESP	134	ESP (SPI=0xf6354fca)
238	184.184840	192.1.1.3	192.1.1.2	ESP	134	ESP (SPI=0xf6354fca)
239	184.188058	192.1.1.3	192.1.1.2	ESP	134	ESP (SPI=0xf6354fca)
240	184.201024	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
241	184.201063	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
242	184.204639	192.1.1.3	192.1.1.2	ESP	166	ESP (SPI=0xf6354fca)
243	184.204937	192.1.1.3	192.1.1.2	ESP	166	ESP (SPI=0xf6354fca)
245	184.560002	192.1.1.3	192.1.1.2	ESP	150	ESP (SPI=0xf6354fca)
246	184.560856	192.1.1.3	192.1.1.2	ESP	134	ESP (SPI=0xf6354fca)

Frame 2: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface en12, id 0
 Ethernet II, Src: Cisco-Afcda61 (00:23:84:af:cd:a6), Dst: Invacast_B5 (01:00:5e:00:00:b5)
 Internet Protocol Version 4, Src: 192.1.1.2, Dst: 224.0.0.5
 Open Connection Data Error

AX88179A: en12: alive capture in progress

Packets: 361, Displayed: 160 (44.3%)

Profile: Default

Ping. connectivity to Server on private RTRA network:

```
(base) loganchayet@econ2-202-233-dhcp ~ % ping 192.168.0.11
PING 192.168.0.11 (192.168.0.11): 56 data bytes
ping: sendto: No route to host
Request timeout for icmp_seq 0
^C
--- 192.168.0.11 ping statistics ---
2 packets transmitted, 0 packets received, 100.0% packet loss
(base) loganchayet@econ2-202-233-dhcp ~ % ping 192.168.0.11
PING 192.168.0.11 (192.168.0.11): 56 data bytes
64 bytes from 192.168.0.11: icmp_seq=0 ttl=254 time=1.654 ms
64 bytes from 192.168.0.11: icmp_seq=1 ttl=254 time=1.545 ms
64 bytes from 192.168.0.11: icmp_seq=2 ttl=254 time=1.491 ms
^C
--- 192.168.0.11 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 1.491/1.563/1.654/0.068 ms
```

RTRA Configuration:

```
aaa authentication login default local
aaa authentication login VPNUSER local
aaa authorization exec default local
aaa authorization network VPNGROUP local
username VPNUSER password 0 CISCO
username admin privilege 15 password 0 admin
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
!
crypto isakmp client configuration group VPNGROUP
```

```
key CISCO
dns 192.168.0.253
wins 192.168.0.253
domain NETWORKLESSONS.LOCAL
pool VPNPOOL
max-users 10
netmask 255.255.255.0
crypto isakmp profile ISAKMPPROFILE
match identity group VPNGROUP
client authentication list VPNUSER
isakmp authorization list VPNGROUP
client configuration address respond
virtual-template 1
!
!
crypto ipsec transform-set TRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile VPNPROFILE
set transform-set TRANSFORMSET
set isakmp-profile ISAKMPPROFILE
!
interface Virtual-Template1 type tunnel
ip unnumbered GigabitEthernet0/0
tunnel mode ipsec ipv4
tunnel protection ipsec profile VPNPROFILE
!
```

Objective 3: DMVPN

1. (Optional) You can either clear configurations from parts 1 and 2 of the first objective or use different tunnel IDs.
2. Consider RTRA as the hub router and R1, R2 and R3 as the spoke routers.
3. Configure the crypto and tunnel on the Spoke router & HUB router, which effectively deploys DMVPN between the Spoke and Hub.
4. Traffic from any branch reaching other branches or the servers inside the company should go to RTRA and then return to the destination via a DMVPN tunnel.

Paste the crypto and NHRP configurations, and various “show” commands to troubleshoot and verify Spoke-Hub connectivity & tunnel formation.

Router Configs:

```
RTRA (R4):
interface tunnel 0
ip address 172.16.123.1 255.255.255.0
tunnel source gi 0/1
```

```
tunnel mode gre multipoint
ip nhrp map multicast dynamic
ip nhrp network-id 1
ip nhrp authentication DMVPN

router eigrp 1
network 192.168.0.0 0.0.0.255
network 172.16.123.0 0.0.0.255
no ip split-horizon eigrp 1

R1:
interface tunnel 0
ip address 172.16.123.2 255.255.255.0
tunnel source gi 0/1
tunnel mode gre multipoint
ip nhrp map 172.16.123.1 192.1.1.2
ip nhrp map multicast 192.1.1.2
ip nhrp network-id 1
ip nhrp nhs 172.16.123.1
ip nhrp authentication DMVPN

router eigrp 1
network 192.168.1.0 0.0.0.255
network 172.16.123.0 0.0.0.255

R2:
interface tunnel 0
ip address 172.16.123.3 255.255.255.0
tunnel source gi 0/1
tunnel mode gre multipoint
ip nhrp map 172.16.123.1 192.1.1.2
ip nhrp map multicast 192.1.1.2
ip nhrp network-id 1
ip nhrp nhs 172.16.123.1
ip nhrp authentication DMVPN

router eigrp 1
network 192.168.2.0 0.0.0.255
network 172.16.123.0 0.0.0.255

R3:
interface tunnel 0
ip address 172.16.123.4 255.255.255.0
tunnel source gi 0/1
tunnel mode gre multipoint
ip nhrp map 172.16.123.1 192.1.1.2
ip nhrp map multicast 192.1.1.2
ip nhrp network-id 1
ip nhrp nhs 172.16.123.1
ip nhrp authentication DMVPN
```

```

router eigrp 1
network 192.168.3.0 0.0.0.255
network 172.16.123.0 0.0.0.255

IPSEC (All routers):
crypto isakmp policy 1
encryption aes
hash md5
authentication pre-share
group 2
lifetime 86400
exit
crypto isakmp key 0 NETWORKLESSONS address 0.0.0.0
crypto ipsec transform-set MYSET esp-aes esp-md5-hmac
exit
crypto ipsec profile MGRE
set security-association lifetime seconds 86400
set transform-set MYSET
exit
interface tunnel 0
tunnel protection ipsec profile MGRE
end

```

Note* OSPF was set up for interface connectivity.

Show commands for verification:

Show dmvpn

```

R4#show dmvpn
Legend: Attrb --> S - Static, D - Dynamic, I - Incomplete
N - NATed, L - Local, X - No Socket
# Ent --> Number of NHRP entries with same NBMA peer
NHS Status: E --> Expecting Replies, R --> Responding, W --> Waiting
UpDn Time --> Up or Down Time for a Tunnel
=====

Interface: Tunnel0, IPv4 NHRP Details
Type:Hub, NHRP Peers:3,

# Ent  Peer NBMA Addr Peer Tunnel Add State  UpDn Tm Attrb
-----
  1    221.1.2.2    172.16.123.2  UP 00:05:36  D
  1    211.1.2.2    172.16.123.3  UP 00:15:22  D
  1    201.1.2.2    172.16.123.4  UP 00:14:41  D
R4#

```

Show ip nhrp

```

R4#show ip nh
R4#show ip nhrp
172.16.123.2/32 via 172.16.123.2
  Tunnel0 created 00:05:59, expire 01:54:01
  Type: dynamic, Flags: unique registered
  NBMA address: 221.1.2.2
172.16.123.3/32 via 172.16.123.3
  Tunnel0 created 00:15:45, expire 01:44:14
  Type: dynamic, Flags: unique registered
  NBMA address: 211.1.2.2
172.16.123.4/32 via 172.16.123.4
  Tunnel0 created 00:15:04, expire 01:44:55
  Type: dynamic, Flags: unique registered
  NBMA address: 201.1.2.2
R4#

```

Show ip route R1, R2, R3, RTRA(R4)

```

R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.123.0/24 is directly connected, Tunnel0
L       172.16.123.2/32 is directly connected, Tunnel0
O       192.1.1.0/24 [110/2] via 221.1.2.1, 00:43:52, GigabitEthernet0/1
D       192.168.0.0/24 [90/26882560] via 172.16.123.1, 00:03:56, Tunnel0
O       192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, Loopback1
L       192.168.1.1/32 is directly connected, Loopback1
D       192.168.2.0/24 [90/28288000] via 172.16.123.1, 00:03:56, Tunnel0
D       192.168.3.0/24 [90/28288000] via 172.16.123.1, 00:03:56, Tunnel0
O       199.1.2.0/24 [110/2] via 221.1.2.1, 00:44:22, GigabitEthernet0/1
O       201.1.2.0/24 [110/2] via 221.1.2.1, 00:44:22, GigabitEthernet0/1
O       211.1.2.0/24 [110/2] via 221.1.2.1, 00:44:22, GigabitEthernet0/1
O       221.1.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       221.1.2.0/24 is directly connected, GigabitEthernet0/1
L       221.1.2.2/32 is directly connected, GigabitEthernet0/1
R1#

```

```

R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.16.123.0/24 is directly connected, Tunnel0
L       172.16.123.3/32 is directly connected, Tunnel0
O       192.1.1.0/24 [110/2] via 211.1.2.1, 00:44:21, GigabitEthernet0/1
D       192.168.0.0/24 [90/26882560] via 172.16.123.1, 00:04:19, Tunnel0
D       192.168.1.0/24 [90/28288000] via 172.16.123.1, 00:04:19, Tunnel0
O       192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, Loopback1
L       192.168.2.1/32 is directly connected, Loopback1
D       192.168.3.0/24 [90/28288000] via 172.16.123.1, 00:04:19, Tunnel0
O       199.1.2.0/24 [110/2] via 211.1.2.1, 00:44:31, GigabitEthernet0/1
O       201.1.2.0/24 [110/2] via 211.1.2.1, 00:44:31, GigabitEthernet0/1
O       211.1.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       211.1.2.0/24 is directly connected, GigabitEthernet0/1
L       211.1.2.2/32 is directly connected, GigabitEthernet0/1
O       221.1.2.0/24 [110/2] via 211.1.2.1, 00:44:31, GigabitEthernet0/1
R2#

```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.123.0/24 is directly connected, Tunnel0
L    172.16.123.4/32 is directly connected, Tunnel0
O    192.1.1.0/24 [110/2] via 201.1.2.1, 00:44:59, GigabitEthernet0/1
D    192.168.0.0/24 [90/26882560] via 172.16.123.1, 00:16:42, Tunnel0
D    192.168.1.0/24 [90/28288000] via 172.16.123.1, 00:04:54, Tunnel0
D    192.168.2.0/24 [90/28288000] via 172.16.123.1, 00:04:54, Tunnel0
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/24 is directly connected, Loopback1
L    192.168.3.1/32 is directly connected, Loopback1
O    199.1.2.0/24 [110/2] via 201.1.2.1, 00:45:19, GigabitEthernet0/1
201.1.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    201.1.2.0/24 is directly connected, GigabitEthernet0/1
L    201.1.2.2/32 is directly connected, GigabitEthernet0/1
O    211.1.2.0/24 [110/2] via 201.1.2.1, 00:44:59, GigabitEthernet0/1
O    221.1.2.0/24 [110/2] via 201.1.2.1, 00:45:19, GigabitEthernet0/1
R3#
```

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

172.16.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.16.123.0/24 is directly connected, Tunnel0
L    172.16.123.1/32 is directly connected, Tunnel0
192.1.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.1.1.0/24 is directly connected, GigabitEthernet0/1
L    192.1.1.2/32 is directly connected, GigabitEthernet0/1
192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.0.0/24 is directly connected, GigabitEthernet0/0
L    192.168.0.1/32 is directly connected, GigabitEthernet0/0
D    192.168.1.0/24 [90/27008000] via 172.16.123.2, 00:08:00, Tunnel0
D    192.168.2.0/24 [90/27008000] via 172.16.123.3, 00:05:21, Tunnel0
D    192.168.3.0/24 [90/27008000] via 172.16.123.4, 00:05:21, Tunnel0
O    199.1.2.0/24 [110/2] via 192.1.1.1, 00:45:23, GigabitEthernet0/1
O    201.1.2.0/24 [110/2] via 192.1.1.1, 00:45:23, GigabitEthernet0/1
O    211.1.2.0/24 [110/2] via 192.1.1.1, 00:45:23, GigabitEthernet0/1
O    221.1.2.0/24 [110/2] via 192.1.1.1, 00:45:23, GigabitEthernet0/1
R4#
```

Show crypto isakmp sa:

```
R4#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
192.1.1.2    221.1.2.2    QM_IDLE        1007 ACTIVE
192.1.1.2    211.1.2.2    QM_IDLE        1005 ACTIVE
192.1.1.2    201.1.2.2    QM_IDLE        1006 ACTIVE

IPv6 Crypto ISAKMP SA
```

Show crypto ipsec sa:

```

R4#show crypto ipsec sa
interface: Tunnel0
  Crypto map tag: Tunnel0-head-0, local addr 192.1.1.2

protected vrf: (none)
local ident (addr/mask/prot/port): (192.1.1.2/255.255.255.255/47/0)
remote ident (addr/mask/prot/port): (201.1.2.2/255.255.255.255/47/0)
current_peer 201.1.2.2 port 500
  PERMIT, flags={origin_is_acl,}
  #pkts encaps: 279, #pkts encrypt: 279, #pkts digest: 279
  #pkts decaps: 277, #pkts decrypt: 277, #pkts verify: 277
  #pkts compressed: 0, #pkts decompressed: 0
  #pkts not compressed: 0, #pkts compr. failed: 0
  #pkts not decompressed: 0, #pkts decompress failed: 0
  #send errors 0, #recv errors 0

local crypto endpt.: 192.1.1.2, remote crypto endpt.: 201.1.2.2
path mtu 1500, ip mtu 1500, ip mtu idb (none)
current outbound spi: 0x430DF1C7(1124987335)
PFS (Y/N): N, DH group: none

inbound esp sas:
  spi: 0x8C2FF122(3157258530)
    transform: esp-aes esp-md5-hmac ,
    in use settings =[{Tunnel,}]
    conn id: 2067, flow_id: Onboard VPN:67, sibling_flags 80000046, crypto map: Tunnel0-head-0
    sa timing: remaining key lifetime (k/sec): (4485013/85262)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE

inbound ah sas:

inbound pcp sas:

outbound esp sas:
  spi: 0x430DF1C7(1124987335)
    transform: esp-aes esp-md5-hmac ,
    in use settings =[{Tunnel,}]
    conn id: 2068, flow_id: Onboard VPN:68, sibling_flags 80000046, crypto map: Tunnel0-head-0
    sa timing: remaining key lifetime (k/sec): (4485012/85262)
    IV size: 16 bytes
    replay detection support: Y
    Status: ACTIVE

outbound ah sas:

outbound pcp sas:

```

Branch C to A traceroute:

```

R3#
R3#traceroute 192.168.1.1
Type escape sequence to abort.
Tracing the route to 192.168.1.1
VRF info: (vrf in name/id, vrf out name/id)
 1 172.16.123.1 4 msec 0 msec 0 msec
 2 172.16.123.2 4 msec * 0 msec
R3#

```

Use the following helpful 'show' commands for verification:

- **show dmvpn**
- **show ip nhrp**
- **show ip route**
- **show crypto isakmp sa**
- **show crypto ipsec sa**

Refer the following link for DMVPN configurations:

<https://www.gns3.com/marketplace/lab/dmvpn-ipsec>